Transformer MLP Transplants Depend on the Receiving State

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Abstract

Activation patching and steering move internal updates between forward passes and often treat a moved update as if it carried a portable effect. Ordinary transplants cannot test this, because the update and the state that receives it change together. We isolate the receiver with a crossed-receiver design: a natural transformer MLP displacement $\Delta u = u_j - u_i$ is computed once, frozen, and physically added to many natural receiver states of Qwen3-8B and Mistral-7B; its exact next-token distributional effect is measured as a KL divergence. The same displacement is mild in some receivers and disruptive in others; across 31 receivers its effect typically spans more than three orders of magnitude. An established receiver-local Fisher score ranks these full-dose effects consistently, with median within-displacement Spearman correlation 0.858–0.928 across all six model–bucket replications; the corresponding unweighted response reaches at most 0.568. The association remains after removing generic displacement strength and receiver fragility, with residual correlation 0.50–0.68. The receiver also changes the direction of the effect: footprints of one displacement diverge across receivers, yet receiver-specific low-rank maps fitted on other displacements predict held-out footprints better than a global map. The effect of an internal update depends on where it is written, in both size and direction.[1]

1 Introduction

Activation patching, path patching, and causal tracing move hidden activations between forward passes to localize the components that drive a behavior [Vig et al., 2020, Geiger et al., 2021, Goldowsky-Dill et al., 2023, Zhang and Nanda, 2024]; activation addition and function vectors move them to control behavior [Turner et al., 2023, Todd et al., 2024]. Both uses read a moved activation as carrying an effect. That reading is only as portable as the downstream network allows, because the remaining layers process the moved activation together with the state that receives it.

Transformer MLPs make the question concrete. At layer ℓ, a host context i and a donor context j produce MLP outputs u_i and u_j. Replacing u_i by u_j is exactly adding the natural displacement $\Delta u_{ij} = u_j - u_i$ to the host's residual stream (Figure 1a). In an ordinary transplant, every host receives a different displacement, so the update and its destination change together. Ordinary transplants can show that host-side quantities are associated with transplant effects (Appendix D.1), but not whether the receiving state changes what a fixed update does.

In this paper, we isolate the receiver. In the crossed-receiver design (Figure 1b, Algorithm 1), a natural displacement is computed once, frozen, and physically added, with unchanged direction and norm, to many other natural receiver states. Because the update is identical across receivers, any variation of its exact effect is caused by the receiving state and its interaction with the update. We

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[1] Code and reproducibility: github.com/mnehan67/neurips-2026

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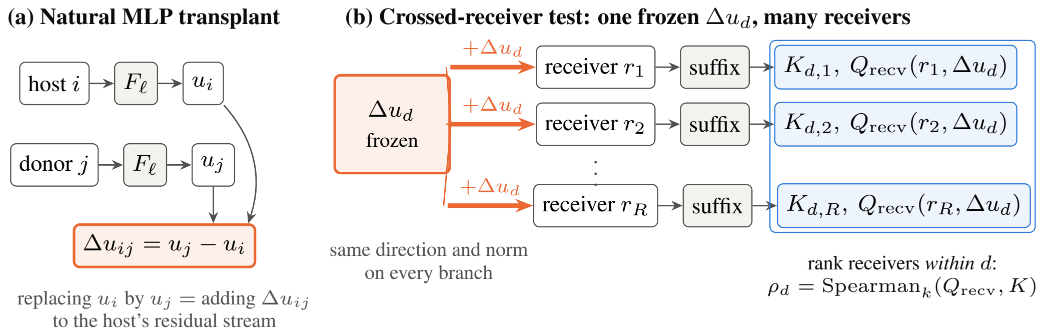
(a) Natural MLP transplant
host i
F_ℓ
u_i
donor j
F_ℓ
u_j
$\Delta u_{ij} = u_j - u_i$
replacing u_i by u_j = adding Δu_{ij} to the host's residual stream
(b) Crossed-receiver test: one frozen Δu_d, many receivers
Δu_d frozen
$+\Delta u_d$
$+\Delta u_d$
$+\Delta u_d$
receiver r_1
receiver r_2
receiver r_R
suffix
suffix
suffix
$K_{d,1},\ Q_{\mathrm{recv}}(r_1, \Delta u_d)$
$K_{d,2},\ Q_{\mathrm{recv}}(r_2, \Delta u_d)$
$K_{d,R},\ Q_{\mathrm{recv}}(r_R, \Delta u_d)$
same direction and norm on every branch
rank receivers *within* d:
$\rho_d = \mathrm{Spearman}_k(Q_{\mathrm{recv}}, K)$

Figure 1: **The crossed-receiver design holds the update fixed and changes only its destination.** (a) Replacing the host MLP output u_i by the donor output u_j adds the natural displacement $\Delta u_{ij} = u_j - u_i$ to the host. (b) In the crossed-receiver test, one natural displacement Δu_d is computed once, frozen, and added unchanged to many natural receiver states $r_1, \ldots, r_R$. For each receiver we measure the exact effect $K_{d,k}$ and the receiver-local score $Q_{\mathrm{recv}}(r_k, \Delta u_d)$, and we rank receivers within the fixed displacement (Algorithm 1).

explain this variation in two steps. For its *size*, we use the receiver-local Fisher score, the standard second-order pullback of output KL through the downstream network [Martens, 2020, Park et al., 2026, Wang and Zhao, 2026], as a diagnostic; we do not claim a new geometry. For its *direction*, we trace the causal footprint of the displacement through every downstream layer and test whether receiver-specific structure in it generalizes to displacements never used for fitting.

What the design adds beyond nonlinearity. A nonlinear network is state dependent in general, so receiver dependence alone is not our claim. We ask how large and how structured it is for natural MLP updates at full dose, with three tests that do not follow from nonlinearity: the primary statistic ranks receivers while the physical update is held fixed; a residual analysis removes generic update strength and receiver fragility before asking whether the Fisher score tracks the pair-specific remainder; and a held-out map test asks whether the receiver-dependent change of direction transfers to unseen displacements, which tests for reusable receiver-specific structure beyond generic state dependence.

Our contributions are as follows (related work is discussed in Appendix A):

- **Crossed-receiver design.** We hold a natural MLP displacement physically fixed and vary only its receiver (Algorithm 1). In Qwen3-8B and Mistral-7B, across three disjoint data buckets and four depths, the exact full-dose effect of a fixed displacement varies strongly and systematically with the receiver (Section 5.2).
- **Receiver-side explanation of effect size.** The established receiver-local Fisher score ranks receivers with median within-displacement $\rho = 0.858$–0.928, ahead of matched-magnitude, attribution-patching, relative-norm, receiver-norm, and alignment baselines (Table 2), and retains a residual association of 0.50–0.68 with pair-specific variation (Table 3).
- **Receiver-conditioned transformation of direction.** The same displacement acquires different downstream causal directions in different receivers, and receiver-specific rank-16 maps predict a reproducible component of this transformation for held-out displacements in both model families (Figure 3).

2 Problem Setup

Fix a frozen decoder-only transformer, an MLP layer ℓ, and contexts of 64 tokens. All interventions act on the MLP output at the final token, and no model parameter is changed.

Natural displacements. Let F_ℓ be the MLP at layer ℓ and let m_c be its input at the final token of context c. A host i and a donor j produce

$$u_i = F_\ell(m_i), \qquad u_j = F_\ell(m_j), \qquad \Delta u_{ij} = u_j - u_i. \tag{1}$$

It is *natural*: both endpoints are MLP outputs the model actually produces.

Algorithm 1 Crossed-receiver test for one natural displacement

```
Require: frozen model; layer $\ell$; host $i$; donor $j$; receivers $k_1, \ldots, k_R$ (natural contexts)
$\Delta u_d \leftarrow F_\ell(m_j) - F_\ell(m_i)$    ▷ natural displacement, computed once
freeze $\Delta u_d$    ▷ identical direction and norm for every receiver
for $k \in \{k_1, \ldots, k_R\}$ do
    run receiver $k$ natively; record $r_k$ and $p_k$
    run receiver $k$ with $r_k \leftarrow r_k + \Delta u_d$; record $p_{k,+\Delta u_d}$
    $K_{d,k} \leftarrow D_{\mathrm{KL}}(p_k \,\|\, p_{k,+\Delta u_d})$    ▷ exact full-dose effect
    record predictors at $(r_k, \Delta u_d)$: $Q_{\mathrm{recv}}$, $q_{p2}$, attribution, norm, and alignment scores
    record causal footprints $c_{d,k}^{(m)} = h_m(k; +\Delta u_d) - h_m(k)$    ▷ control panel only
end for
$\rho_d \leftarrow \mathrm{Spearman}_{k \neq i}(\text{predictor}_{d,k},\ K_{d,k})$    ▷ update fixed inside $\rho_d$
return $\rho_d$; the primary statistic is $\mathrm{median}_d\, \rho_d$ over displacements
```

Receivers and the exact effect. For any context k, let r_k be the residual state immediately after the layer-ℓ MLP at the final token, and let p_k be the native next-token distribution. Writing a displacement Δu into receiver k replaces r_k by $r_k + \Delta u$ (equivalently, the MLP output u_k by $u_k + \Delta u$) and leaves the rest of the forward pass unchanged. With $p_{k,+\Delta u}$ the resulting next-token distribution, the exact next-token distributional effect (for short, the exact effect) is

$$K_k(\Delta u) = D_{\mathrm{KL}}\big(p_k \,\big\|\, p_{k,+\Delta u}\big)\,. \tag{2}$$

The forward KL from the native distribution, over the full vocabulary, measures how far the physically applied intervention moves the model from its unpatched behavior. An ordinary transplant is the special case $K_{i\leftarrow j} = K_i(\Delta u_{ij})$; a self-patch recovers the native output to numerical precision.

Goal. Our goal is to determine whether, for a fixed natural displacement Δu_d, the effect $K_k(\Delta u_d)$ depends on the receiver k, how strongly it does so, and whether receiver-side quantities explain both the size and the direction of the dependence.

3 The Crossed-Receiver Design

We first present the protocol and its statistics; the receiver-side quantities it records are defined in Section 4.

3.1 Protocol

Each crossed experiment samples natural displacements $\Delta u_d = u_{j_d} - u_{i_d}$ from host–donor pairs and crosses each with a set of natural receivers, never rescaling or re-orienting it (Algorithm 1). The original host is kept as an anchor row; the primary statistic uses only alternative receivers $k \neq i_d$.

3.2 Three correlations, one primary statistic

The protocol yields displacement-by-receiver matrices of exact effects $K_{d,k}$ and predictor values $X_{d,k}$. We summarize their association in three ways that answer three different questions (formal definitions in Appendix B.4).

1. **Pooled correlation** $\rho_{\mathrm{pool}} = \mathrm{Spearman}(X_{d,k}, K_{d,k})$ over all rows. It is descriptive and mixes update, receiver, and pair-specific variation.
2. **Within-displacement correlation (primary)** $\mathrm{median}_d\, \rho_d$, where ρ_d correlates $X_{d,k}$ and $K_{d,k}$ across the receivers of one fixed displacement. It asks whether X ranks receivers correctly *while the physical update is held fixed.*
3. **Residual-rank correlation.** Rank-transform X and K, remove additive displacement and receiver main effects from each by alternating demeaning, and correlate the residuals. It asks whether X still captures pair-specific structure *after generic displacement strength and receiver fragility are removed.* It was introduced after inspecting the crossed data and is reported as post hoc.

These are not competing estimates of one quantity. In the six crossed matrices, receiver identity alone explains 0.70–0.80 of the rank variance of exact KL, displacement identity 0.04–0.12, and 0.18–0.25 remains as non-additive residual variation (Table 4). A high within-displacement correlation thus largely means that a predictor identifies generically fragile receivers; the residual correlation asks whether it also tracks the smaller non-additive part. Values near 0.9 for the first question and 0.5–0.68 for the second are therefore consistent. With one effect per cell, we do not interpret the remainder as a separately identified interaction. Every Fisher–KL correlation in the paper is one of these types on a specific panel; Table 5 lists them side by side.

What the design identifies. Within a displacement the intervention vector is physically identical across receivers, so differences in $K_{d,k}$ are caused by the receiving state and its interaction with the update, for the given model and layer. The design does not identify which mechanism (attention, normalization, or MLP gating) produces the dependence, nor whether an MLP update represents a semantic function; our endpoints are next-token behavior and causal-effect geometry.

4 Receiver-Side Measurements

4.1 Receiver-local sensitivity

Let G_ℓ map the post-MLP residual state at layer ℓ to the final logits, with the rest of the context held fixed. A small movement of receiver r along Δu changes the logits by $\Delta z = JG_\ell(r)\,\Delta u$. With p the receiver's native next-token distribution and $C_p = \mathrm{diag}(p) - pp^\top$, the receiver-local score is

$$Q_{\mathrm{recv}}(r, \Delta u) = \tfrac{1}{2}\left(JG_\ell(r)\Delta u\right)^\top C_p \left(JG_\ell(r)\Delta u\right). \tag{3}$$

This is the standard second-order pullback of output KL, $D_{\mathrm{KL}}(p_r \,\|\, p_{r+\epsilon\Delta u}) = \epsilon^2\, Q_{\mathrm{recv}}(r, \Delta u) + O(\epsilon^3)$ [Martens, 2020, Park et al., 2026, Wang and Zhao, 2026] (Appendix B.1); we use it as a diagnostic rather than claim a new geometry. The empirical question is whether this local quantity, evaluated at each receiver, ranks the *full-dose* effect ($\epsilon = 1$) of a fixed natural displacement.

Remark 1 (Standard variance reading). $(JG_\ell\Delta u)^\top C_p\,(JG_\ell\Delta u) = \mathrm{Var}_{y\sim p}\big[(JG_\ell\Delta u)_y\big]$, so Q_{recv} is half the variance of the logit change under the receiver's own next-token distribution. A uniform logit shift costs nothing, and changes on tokens the receiver considers unlikely cost little. The matched baseline $q_{p2} = \frac{1}{2}\|JG_\ell(r)\Delta u\|_2^2$ uses the same response without this weighting, isolating the contribution of the receiver's output distribution.

The crossed experiments estimate $JG_\ell(r)\Delta u$ by centered finite differences ($\epsilon = 0.075$) on FP16 model evaluations. An exact FP32 downstream-suffix JVP audit on 16 control-panel cases per model shows negligible truncation error (median cosine 1.0000; $\rho = 1.000$ between FP32 finite-difference and exact Fisher scores), whereas FP16 scores for the same cases agree less closely ($\rho = 0.785$, 0.544; Appendix D.12). We therefore use the FP16 score as a ranking diagnostic: the primary crossed result is an empirical association between this computed score and the independently measured full-dose KL, and it does not require the score to be a numerically exact FP32 Fisher quadratic. The directional results (Sections 5.5 and 5.6) do not use the score.

Alternative predictors. We compare Q_{recv} with q_{p2}; with first-order attribution of the native top-token log-probability (AtP-top) and of the native top1–top2 logit margin (AtP-margin), in the spirit of attribution patching [Kramár et al., 2024]; with the relative displacement norm $\|\Delta u\|/\|r\|$; with the receiver norm $\|r\|$; and with the alignment $|\cos(\Delta u, r)|$. We also run a physical control in which every natural displacement is rescaled to a common norm before being applied (Appendix C.5).

Remark 2 (Why KL is not attributed directly). With p_α the distribution after adding $\alpha\Delta u$, $\frac{d}{d\alpha} D_{\mathrm{KL}}(p \,\|\, p_\alpha)\big|_{\alpha=0} = 0$ (Appendix B.3), so first-order attribution of KL is identically zero and AtP uses the non-stationary targets above, in absolute value.

4.2 Following the causal effect downstream

KL measures how much the final distribution changes, but not the direction of the hidden-state effect. For displacement d with original host i and receiver k, the causal footprint at downstream layer m is

$$c_{d,k}^{(m)} = h_m(k; +\Delta u_d) - h_m(k), \tag{4}$$

Table 1: **Experimental panels.** All panels use frozen weights, 64-token C4 validation contexts, and final-token MLP interventions. "Rows" counts interventions per depth, per bucket, or per Pythia run (256 hosts); a depth count of 1 means the primary depth (layer 2 for Pythia). Crossed receiver counts include the original host as an anchor row; the within-displacement statistic uses the 15 or 31 alternative receivers.

Panel	Design	Rows	Depths	Used for
Ordinary	all directed pairs, 64 contexts	4032	4	residual; wrong-state; dose; horizon
Large ordinary	Mistral, 256 contexts	65,280	1	scale; permutation test
Crossed B1	512 Δu × 16 receivers	8192	4	primary statistic; depth
Crossed B2, B3	512 Δu × 32 receivers	16,384	1	replication
Control	96 Δu × 8 receivers	768	1	baselines; footprints; maps
Pythia-70M (×6)	ordinary, 5-NN donors	256	1	independent pretraining

the difference between patched and native final-token hidden states, with $c_{d,i}^{(m)}$ defined analogously at the original host. Both footprints equal Δu_d at the intervention site. We summarize how they separate with the directional fidelity $F_{d,k}^{(m)} = \cos(c_{d,i}^{(m)}, c_{d,k}^{(m)})$, the gain $A_{d,k}^{(m)} = \langle c_{d,i}^{(m)}, c_{d,k}^{(m)} \rangle / \|c_{d,i}^{(m)}\|_2^2$, and the rewrite residual $E_{d,k}^{(m)} = c_{d,k}^{(m)} - A_{d,k}^{(m)} c_{d,i}^{(m)}$. Together they distinguish an effect that is preserved but attenuated or amplified ($F \approx 1$, $A \neq 1$) from one whose direction is rewritten (small F, large E).

Held-out receiver maps. To test whether the receiver-dependent transformation has reusable structure, we fit at each downstream layer a low-rank map $T_k^{(m)}$ predicting $c_{d,k}^{(m)}$ from $c_{d,i}^{(m)}$. Displacements are split once, before analysis, into 75% training and 25% held out; a rank-16 PCA basis is fit on training displacements only, and ridge maps are fit in that subspace. We compare five predictors of the held-out footprint: the *identity* ($c_k \approx c_i$), a *gain-only* scalar per receiver, one *global map* shared by all receivers, the *receiver-specific map*, and a *wrong-receiver map* taken from another receiver by a fixed cyclic permutation. If receiver identity carried no reusable structure beyond a shared transformation, receiver-specific maps should not systematically outperform a global map on held-out displacements; under a pure change of scale, gain only would suffice. We score cosine and normalized RMSE on held-out displacements, with the held-out displacements as independent units after averaging receivers (Appendix B.5).

5 Experiments

We apply the design to two 7–8B models, then test the size explanation against baselines and pair-specific structure, and finally the direction of the effect and its generalization.

5.1 Setup

Models. We use Qwen3-8B [Yang et al., 2025] (36 layers) and Mistral-7B-v0.3 [Jiang et al., 2023] (32 layers) with frozen FP16 weights. The four probed depths are Qwen layers 7, 14, 21, 26 and Mistral layers 6, 12, 19, 23; the primary depths, at about 40% of each network, are Qwen layer 14 and Mistral layer 12. Six independently pretrained Pythia-70M runs [Biderman et al., 2023, van der Wal et al., 2025] provide a separate ordinary-transplant replication (Section 5.7).

Data. Contexts are deterministic samples from the C4 validation split [Raffel et al., 2020], selected by a SHA-256 hash rule before model-specific tokenization. Each context has 64 tokens, and every intervention and measurement uses the final token. The three crossed buckets B1–B3 are assigned by distinct hash residues and are therefore disjoint.

Panels. Table 1 lists the panels. The crossed buckets carry the primary result. The control panel is a separate deterministic 96 × 8 panel, sampled with a different seed and selection rule; it stores layerwise hidden-state differences for the baselines, footprints, and held-out maps, and its 96 displacements are split once into 72 training and 24 held-out. The primary statistic and baselines are those of Sections 3.2 and 4.1.

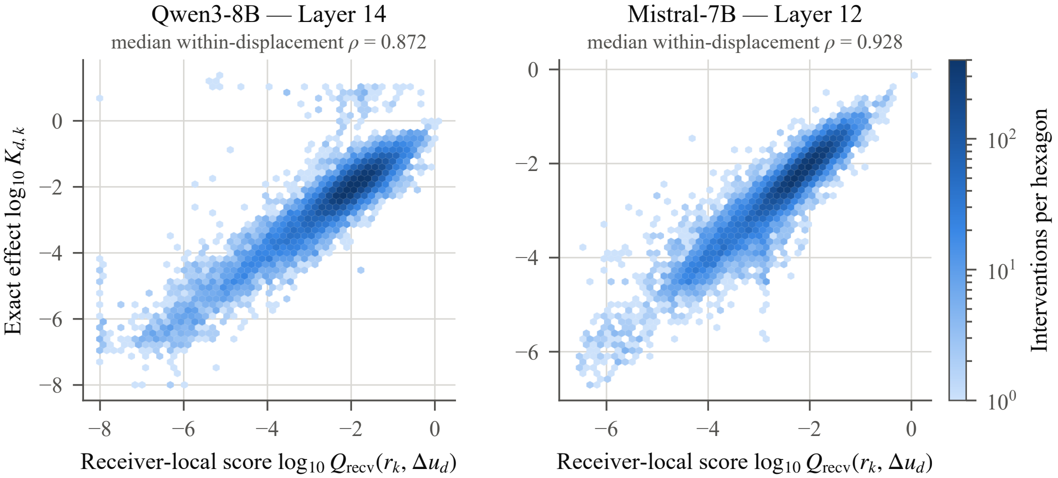

Figure 2: **Receiver-local sensitivity ranks the effects of a fixed displacement across receivers.** Bucket B2, primary depth: 512 frozen natural displacements, each applied to 31 alternative receivers (15,872 interventions per model); both axes $\log_{10}$. Headers give the primary statistic. Other buckets and depths: Tables 3 and 6.

Table 2: **Alternative predictors of receiver ordering.** Median within-displacement Spearman correlation (↑) with exact full-dose KL on the 96×8 control panel; bold marks the best per model. With 8 receivers per displacement, medians are coarser and lower than in the 32-receiver buckets (Table 5); pooled and residual versions are in Table 8.

Model	Fisher Q_{recv}	AtP-top	AtP-margin	q_{p2}	$\lVert\Delta u\rVert/\lVert r\rVert$	$\lVert r\rVert$	$\lvert\cos(\Delta u, r)\rvert$
Qwen3-8B	**0.738**	0.500	0.429	0.226	0.071	−0.071	−0.012
Mistral-7B	**0.905**	0.750	0.179	0.060	0.000	0.000	0.048

Implementation details. We use Hugging Face Transformers and shard the large models with Accelerate across two NVIDIA T4 GPUs (16 GB each). No model is trained or fine-tuned. Seeds, context identifiers, and the split are fixed before analysis. Protocols, compute, and frozen outputs are in Appendix C and at `https://anonymous.4open.science/r/newinml-7402/`.

5.2 The same displacement has different effects in different receivers

A fixed natural displacement does not have one effect. With 31 alternative receivers (buckets B2 and B3), the largest and smallest exact effects of the same displacement differ by a median factor of $10^{4.4}$–$10^{4.8}$ in Qwen3-8B and $10^{3.3}$–$10^{3.6}$ in Mistral-7B, and by at least a factor of 100 for 99.0–100% of displacements (Appendix D.5).

This variation is ordered. The receiver-local score ranks receivers with median within-displacement correlation $0.871/0.872/0.858$ in Qwen3-8B buckets B1/B2/B3 and $0.911/0.928/0.928$ in Mistral-7B (Figure 2), and the correlation is positive for 99.4–99.8% of Qwen displacements and every sampled Mistral displacement. The unweighted response q_{p2}, computed from the same finite differences, reaches only 0.340–0.355 in Qwen and 0.510–0.568 in Mistral, so weighting by the receiver's own next-token distribution (Remark 1) accounts for much of the ranking.

5.3 Scale, attribution, and alignment baselines do not explain the ordering

On the control panel (Table 2), the Fisher score is the strongest predictor in both models. Attribution of the native top-token log-probability is the closest competitor (0.500 in Qwen3-8B, 0.750 in Mistral-7B); the relative displacement norm, receiver norm, and displacement–state alignment are near zero. A physical control gives the sharpest scale check: rescaling every natural displacement to a common norm before applying it leaves the receiver ordering of exact KL unchanged (median within-displacement $\rho = 1.000$ between original and equal-norm orderings in both models). Since a

Table 3: **Within-displacement and pair-specific association in the six crossed replications.** "Within" is the primary statistic. "Residual" is the post-hoc rank correlation after removing additive displacement and receiver effects (Section 3.2). "Non-additive" is the share of rank-KL variance not explained by the additive displacement-plus-receiver model (Table 4). Higher is better for all correlation columns.

		Within-displacement ρ		Residual ρ		
Model	Bucket	Fisher Q_{recv}	q_{p2}	Fisher Q_{recv}	q_{p2}	Non-additive
Qwen3-8B	B1	0.871	0.355	0.504	0.284	0.211
Qwen3-8B	B2	0.872	0.340	0.504	0.292	0.221
Qwen3-8B	B3	0.858	0.352	0.520	0.317	0.246
Mistral-7B	B1	0.911	0.529	0.664	0.411	0.196
Mistral-7B	B2	0.928	0.510	0.675	0.442	0.185
Mistral-7B	B3	0.928	0.568	0.658	0.414	0.180

displacement's scale factor is shared by all its receivers, displacement norm does not produce the receiver ordering.

5.4 Pair-specific structure remains after removing generic fragility

A high within-displacement correlation could reflect generic receiver fragility alone. Table 3 separates the two: 18–25% of the rank variance of exact KL is non-additive in every crossed matrix, and after removing additive displacement and receiver effects the Fisher score's residual correlation is 0.504–0.520 in Qwen3-8B and 0.658–0.675 in Mistral-7B, above q_{p2} in all six replications (0.284–0.442). The score therefore tracks pair-specific variation, not only fragility; the analysis is post hoc and descriptive.

5.5 The receiver changes the direction of the causal effect

The scalar results leave open whether receivers merely amplify or attenuate a fixed update. On the control panel, the footprints of one displacement at the original host and at a receiver start identical and then separate quickly: median directional fidelity falls from 1 to 0.030 at Qwen3-8B's final layer and 0.037 at Mistral-7B's, and the median gain decays with it (Figure 4). The receiver's final footprint is nearly orthogonal to the host's, so the receiver rewrites the direction of the effect rather than rescaling a shared one.

5.6 A receiver-specific transformation generalizes to held-out displacements

Figure 3 shows that part of this rewriting is systematic. At the final layer, receiver-specific maps reach mean held-out cosine 0.316 in Qwen3-8B and 0.241 in Mistral-7B, against 0.086 and 0.109 for the global map and -0.023 and 0.028 for wrong-receiver maps. Bootstrapping the 24 held-out displacements, the advantage over the global map is $+0.230$ (95% CI $[0.205, 0.256]$) in Qwen3-8B and $+0.132$ ($[0.108, 0.155]$) in Mistral-7B, and over wrong-receiver maps $+0.340$ ($[0.281, 0.400]$) and $+0.213$ ($[0.180, 0.244]$): the predictive structure belongs to the specific receiver. Exact vector reconstruction improves much less (normalized RMSE 0.945 versus 1.005 for the global map in Qwen3-8B, 0.986 versus 0.999 in Mistral-7B; Table 9). The transformation thus contains a reproducible receiver-specific *component*, while substantial update-specific variation remains.

5.7 Robustness

Depth and scale. In the B1 depth sweep, the median within-displacement correlation ranges from 0.857 to 0.954 across the four Qwen depths and from 0.854 to 0.957 across the four Mistral depths (Table 6). A 65,280-pair Mistral matrix of ordinary transplants gives Fisher–KL correlation 0.911 and host–donor residual correlation 0.660 (q_{p2}: 0.415), with joint context-label permutation $p = 0.0005$, the smallest value attainable with 2000 permutations (Appendix D.1).

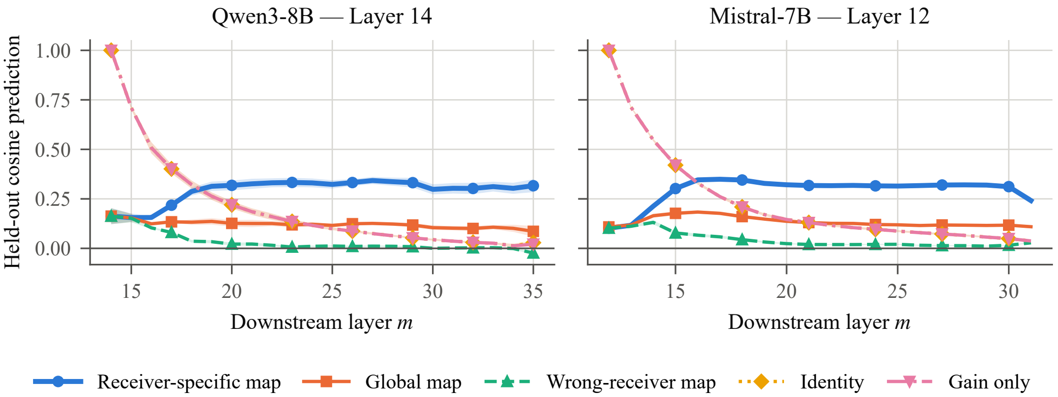

Figure 3: **A receiver-specific transformation predicts the footprints of unseen displacements.** Mean cosine between predicted and actual held-out footprints $c_{d,k}^{(m)}$ at every downstream layer m, on the 96×8 control panel. Maps are fit on 72 natural displacements and evaluated on 24 held-out displacements. Bands: ± 1 standard error across the 24 held-out displacements, after averaging the eight receivers within each. Identity and gain only coincide (a scalar gain does not change direction) and decay with depth; receiver-specific rank-16 maps keep signal through the final layer, global maps are weaker, and wrong-receiver maps fall to about zero.

Dose and horizon. On a random 256-pair subset of the ordinary matrices, the score stays informative up to the full transplant (Fisher–KL correlation 0.831 in Qwen3-8B and 0.907 in Mistral-7B at dose $\lambda = 1$; Figure 5). On a random 128-pair subset run on the true continuation, receiver-local sensitivity remains associated with KL summed over the next h teacher-forced positions, although the association weakens with the horizon ($0.857 \rightarrow 0.565$ in Qwen3-8B and $0.930 \rightarrow 0.812$ in Mistral-7B from $h = 1$ to 32; Appendix D.10). Teacher forcing does not establish persistence under unconstrained autoregressive generation.

Independent pretraining. On six independently pretrained Pythia-70M runs, all six partial Spearman correlations between $\log(1 + Q_{\text{recv}})$ and transplant KL, controlling for update and state distance, nonlinear residual, and tangent alignment, are positive (median 0.800, 95% bootstrap interval $[0.742, 0.827]$; Appendix D.11). This uses ordinary transplants, not the crossed design: it shows that the local-geometry association replicates across independently pretrained small models, while the crossed-receiver evidence comes from the two large models.

6 Conclusion

A natural MLP displacement does not carry one context-free behavioral effect. Holding the update physically fixed while changing only its receiver produces large, systematic differences in exact next-token behavior, which an established receiver-local Fisher score ranks beyond generic receiver fragility and beyond scale, attribution, and alignment baselines. The receiver also rewrites the downstream direction of the effect, and receiver-specific maps recover a reproducible component of that transformation for unseen displacements. Analyses that move internal updates across contexts should treat the receiving state as part of the intervention.

Limitations. We study final-token MLP outputs and next-token behavior; other components and longer horizons are not covered. The large-scale evidence comes from one Qwen3-8B and one Mistral-7B checkpoint, and independent-pretraining replication exists only for Pythia-70M under ordinary transplants. The residual analysis is post hoc and descriptive. The receiver maps are linear and rank-16 and measure causal-effect geometry, not preservation of a semantic function. The crossed experiments use FP16 finite-difference Fisher scores; the FP32 suffix audit closes truncation error on the tested directions but does not make these scores numerically exact (Appendix D.12). Teacher-forced checks do not establish behavior under free generation. The study is diagnostic and proposes no steering method; preserving an intervention's intended causal effect across receiver states is a natural direction for future work (Appendix E).

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A Related Work

Activation transplantation and causal interventions. Activation patching, interchange interventions, causal tracing, path patching, and circuit discovery intervene on hidden states to identify the components that causally drive a behavior [Vig et al., 2020, Geiger et al., 2021, 2025, Goldowsky-Dill et al., 2023, Conmy et al., 2023, Wu et al., 2023, Zhang and Nanda, 2024]. AtP* approximates exhaustive patching with gradients [Kramár et al., 2024]. Patchscopes, Future Lens, function vectors, relational-concept extraction, and concept transplantation move information or behavior across contexts [Ghandeharioun et al., 2024, Pal et al., 2023, Todd et al., 2024, Wang et al., 2024, Dong et al., 2025]. Transfer is not automatic: the outcome of activation swaps depends on the protocol, several mediators can interact, causal vectors need not be invariant across formats, intervened representations can diverge from the model's natural distribution, and role claims about components can fail to transfer [Garcia, 2026, Vaidyanathan et al., 2026, Opiełka et al., 2026, Grant et al., 2026, Quirke, 2026]. Our design differs from these lines of work by holding an example-specific natural MLP displacement exactly fixed while changing only the natural state that receives it, so that the receiver's causal contribution is isolated.

MLP updates as units of computation. Feed-forward layers have been described as key–value memories and as promoting concepts in vocabulary space, and editing their weights is a standard way to change stored associations [Geva et al., 2021, 2022, Meng et al., 2022]. These views treat an MLP output as carrying content. We do not assign content to an MLP update; we measure how its causal effect depends on where it is written.

State-dependent steering and local geometry. Activation steering modifies hidden representations to change behavior [Turner et al., 2023, Rimsky et al., 2024, Li et al., 2023, Zou et al., 2023, Singh et al., 2024]. Recent work makes state dependence explicit through context-conditioned steering fields, local linear control, and response-theoretic sensitivity [Li et al., 2026, Skifstad et al., 2026, Olivieri and Pérez Rodríguez, 2026]. The softmax output has a natural Fisher geometry tied to behavioral change [Martens, 2020, Park et al., 2026], and FishBack pulls this geometry back to intermediate transformer states through the downstream Jacobian in order to steer [Wang and Zhao, 2026]. We do not propose a new geometry or a steering method. We use the established local metric as a diagnostic for whether the same natural update has different full-dose effects at different receivers.

Reusable internal computation. Hidden APIs asks whether hidden interfaces support reusable causal computations across forked futures [Ma et al., 2026]. Function vectors and related representation-transfer studies test whether particular behaviors remain portable across prompts or formats [Todd et al., 2024, Opiełka et al., 2026, Quirke, 2026]. Our transformation analysis does not assign a semantic function to an MLP update. It defines the computation operationally by its measured causal footprint and asks whether receiver-specific structure in that footprint generalizes to unseen natural displacements.

B Mathematical Details

The identities in this appendix are standard. We state them to make the diagnostic and the baselines precise, not as new results.

B.1 Local KL identity

Let $A(z) = \log \sum_y e^{z_y}$, so that $\nabla A(z) = p = \operatorname{softmax}(z)$ and $\nabla^2 A(z) = C_p = \operatorname{diag}(p) - pp^\top$. For logits z and a logit change Δ,

$$D_{\mathrm{KL}}\big(\operatorname{softmax}(z) \,\|\, \operatorname{softmax}(z+\Delta)\big) = A(z+\Delta) - A(z) - \langle p, \Delta\rangle = \tfrac{1}{2}\,\Delta^\top C_p\,\Delta + O(\|\Delta\|^3). \quad (5)$$

Adding $\epsilon\Delta u$ to the receiver state gives $\Delta(\epsilon) = G_\ell(r + \epsilon\Delta u) - G_\ell(r) = \epsilon\, JG_\ell(r)\Delta u + O(\epsilon^2)$. Because the first-order term of (5) vanishes, the $O(\epsilon^2)$ part of Δ enters the divergence only at order ϵ^3, and

$$D_{\mathrm{KL}}(p_r \,\|\, p_{r+\epsilon\Delta u}) = \epsilon^2\, Q_{\mathrm{recv}}(r, \Delta u) + O(\epsilon^3), \quad (6)$$

with Q_{recv} as in (3). This is the classical second-order expansion of KL in a curved exponential family [Martens, 2020] pulled back through the downstream map [Park et al., 2026, Wang and Zhao, 2026]. The experiments evaluate the score at $\epsilon = 1$, where this expansion is not guaranteed to be accurate; whether it still ranks receivers is an empirical question.

B.2 Variance form and the matched baseline

For any $v \in \mathbb{R}^V$, $v^\top C_p v = \sum_y p_y v_y^2 - \left(\sum_y p_y v_y\right)^2 = \mathrm{Var}_{y\sim p}[v_y]$, which gives Remark 1. In particular $C_p \mathbf{1} = 0$, so a uniform logit shift has zero score. Both Q_{recv} and the matched baseline

$$q_{p2}(r, \Delta u) = \tfrac{1}{2} \| J G_\ell(r) \Delta u \|_2^2 \tag{7}$$

use the same finite-difference estimate of $JG_\ell(r)\Delta u$; they differ only in the weighting by the receiver's next-token distribution.

B.3 First-order attribution of KL is zero

Let z_α be the logits after adding $\alpha \Delta u$, with $z_0 = z$. Differentiating (5) with $\Delta = z_\alpha - z$ gives $\frac{d}{d\alpha} D_{\mathrm{KL}}(p \,\|\, p_\alpha) = \langle p_\alpha - p, \dot{z}_\alpha \rangle$, which is zero at $\alpha = 0$. A first-order attribution score for KL is therefore identically zero at the native point. The AtP baselines instead use two non-stationary targets: the directional derivative of the native top-token log-probability and the directional derivative of the native top1–top2 logit margin. We take absolute values because the question is how strongly a receiver reacts along the intervention direction.

B.4 The three correlation estimands

Let $\mathcal{K}_d$ be the receivers of displacement d, including the original host i_d as an anchor, and let $X_{d,k}$ be a predictor.

1. **Pooled:** $\rho_{\mathrm{pool}} = \mathrm{Spearman}\big(\{X_{d,k}\}, \{K_{d,k}\}\big)$ over all rows, anchors included.
2. **Within-displacement (primary):** $\rho_d = \mathrm{Spearman}_{k \in \mathcal{K}_d \setminus \{i_d\}}(X_{d,k}, K_{d,k})$, reported as $\mathrm{median}_d\, \rho_d$ together with the fraction of displacements with $\rho_d > 0$. Displacements with fewer than four alternative receivers would be skipped; none are in the reported panels.
3. **Residual-rank (crossed matrices):** rank-transform X and K over all rows and standardize them, then remove displacement and receiver means by alternating demeaning until convergence. The statistic is the Pearson correlation of the two residual vectors.

For the ordinary host–donor matrices (Appendix D.1), the pair-specific statistic is computed in the same spirit, but on the original scale: $\log(1 + \cdot)$-transformed values are two-way demeaned by host and donor, and the residuals are compared by Spearman correlation. We report each statistic with the method used to compute it in the frozen artifact; the two procedures are not numerically interchangeable.

B.5 Causal footprints and held-out receiver maps

For downstream layer m, the footprints $c_{d,i}^{(m)}$ and $c_{d,k}^{(m)}$ are differences between patched and native final-token hidden states (Equation (4)). Fidelity, gain, and the rewrite residual are defined in Section 4.2. For prediction, the 96 control-panel displacements are split once into 72 training and 24 held-out displacements, and the split is fixed before analysis. At each layer, a rank-16 PCA basis is fit to training footprints only, and ridge maps (regularization strength 1.0) are fit in this subspace. The baselines are the identity; a scalar gain per receiver; one global map using all receivers; the receiver-specific map; and a wrong-receiver map obtained by a fixed cyclic permutation of receiver identities. We report cosine prediction and normalized RMSE on held-out displacements. For uncertainty, receiver rows are averaged within each held-out displacement, so the 24 held-out displacements are the independent units for both the standard-error bands in Figure 3 and the bootstrap intervals in Section 5.6.

C Experimental and Reproducibility Details

C.1 Models, data, and intervention sites

We use Qwen3-8B and Mistral-7B-v0.3 with frozen weights. Contexts are sampled deterministically from the C4 validation split with SHA-256 thinning before model-specific tokenization. Each context contains 64 tokens, and all interventions and measurements use the final token. The four depth probes are Qwen layers 7, 14, 21, 26 and Mistral layers 6, 12, 19, 23, at relative depths of about 0.20, 0.40, 0.60, and 0.74. The primary depths are Qwen layer 14 and Mistral layer 12. Exact checkpoint revisions are recorded in the artifact.

C.2 Ordinary transplant protocol

The ordinary protocol uses 64 contexts and evaluates every directed non-self donor–host pair, which gives $64 \times 63 = 4032$ interventions at each tested depth. It supports host–donor residualization, the wrong-state diagnostic, and comparison with earlier activation-transplant settings. It does not isolate receiver causality by itself, because the displacement changes with the host. At the primary depth, two random subsets of this matrix are reused: 256 pairs for the dose curve (Figure 5) and 128 pairs for the teacher-forced check (Appendix D.10). A larger ordinary matrix covers all $256 \times 255 = 65{,}280$ directed pairs of the 256 contexts of bucket B1 in Mistral-7B at layer 12.

C.3 Crossed-receiver protocol

The crossed-receiver test uses three disjoint 256-context buckets, assigned by distinct SHA-256 residues modulo 11. Each stage samples 512 natural displacements from host–donor pairs within the bucket. In bucket B1, each displacement is crossed with 16 receivers (the original host as an anchor plus 15 alternatives), giving 8192 interventions per stage. B1 is repeated at all four depths in both models. Buckets B2 and B3 use 32 receivers (the anchor plus 31 alternatives) at the primary depth, giving 16,384 interventions each. In total, the twelve crossed stages contain 131,072 physical interventions. The anchor rows enter the pooled and residual statistics and the variance decomposition, but not the within-displacement statistic.

C.4 Control panel

The control panel is a separate deterministic 96×8 panel per model at the primary depth: 96 natural displacements, built from disjoint host and donor contexts, each crossed with the same eight receivers. Its contexts are selected by a different hash rule from the crossed buckets, and we do not assume that it is statistically independent of them. The panel stores layerwise final-token hidden-state differences, which the scalar-only crossed buckets do not. It is used for the alternative predictors, the equal-norm control, the causal footprints, and the held-out maps.

C.5 Alternative predictors and the equal-norm control

For every row of the control panel, we record Q_{recv}, q_{p2}, the two AtP scores (Appendix B.3), the receiver norm $\|r\|$, the relative displacement norm $\|\Delta u\|/\|r\|$, and $|\cos(\Delta u, r)|$. The AtP gradients are exact automatic-differentiation gradients of the respective targets with respect to the receiver state, with one backward pass through the suffix per receiver. For the physical equal-norm control, each natural displacement is rescaled to the median displacement norm of the panel before it is applied to its receivers, and exact KL is recomputed. Because a displacement's scale factor is shared by all of its receivers, the control asks whether the nonlinear receiver ordering changes once between-displacement magnitude differences are removed.

C.6 Implementation and compute

The large-model experiments use Hugging Face Transformers with FP16 weights, sharded with Accelerate across two NVIDIA T4 GPUs (16 GB each), with gradients disabled except for the AtP baselines, 64-token contexts, and pair batches of four. The twelve crossed-receiver stages and the 65,280-pair Mistral matrix have instrumented stage runtimes totalling 631.3 minutes (437.8 minutes crossed, 193.4 minutes large matrix). The eight ordinary 64-context depth targets total 134.9 minutes, as recorded in the artifact. The wall-clock time of the two control-panel runs was not instrumented, so we do not report it. The Pythia runs use FP32 and exact JVPs (Appendix D.11). The numerical audit of Appendix D.12.2 produces the natural states with the FP16 model on the same two T4 GPUs and then evaluates the downstream suffix in FP32, with TF32 disabled, split across both GPUs; its runtime was not recorded. Random seeds, context identifiers, and train/held-out splits are fixed before analysis. Frozen outputs, analysis scripts, and figure-generation code are available at `https://anonymous.4open.science/r/newinml-7402/`. All data figures in this paper are regenerated from these frozen outputs. Row-level outputs allow every crossed-receiver statistic and the Mistral control-panel intervals to be recomputed on CPU. For the Qwen3-8B control panel, the artifact stores the bootstrap and standard-error summaries but not per-displacement map scores.

D Additional Results

D.1 Ordinary transplants and the wrong-state diagnostic

At the primary depth, the ordinary-transplant Fisher–KL Spearman correlation over all 4032 pairs is 0.867 in Qwen3-8B and 0.911 in Mistral-7B. Removing additive host and donor effects leaves 0.563 and 0.677, with two-way bootstrap intervals $[0.484, 0.637]$ and $[0.607, 0.750]$. The wrong-state diagnostic keeps each transplant outcome fixed but evaluates the score of its displacement at unrelated states. On a random 256-pair subset, the correlation between the transplant KL and the score at the correct host is 0.875 in Qwen3-8B and 0.919 in Mistral-7B. When the score is evaluated at eight unrelated states per pair, it drops to -0.015 and 0.028 (2048 rows). The diagnostic shows that the score must be computed at the state that receives the update. It does not show that the receiving state causes the variation, because the displacement still changes with the host; the crossed design in the main text tests that directly.

For the larger Mistral matrix of 65,280 ordinary transplants, the raw Fisher–KL correlation is 0.911, the host–donor residual correlation is 0.660, and the corresponding q_{p2} residual correlation is 0.415. A joint context-label permutation test, which applies the same permutation to both matrix axes, gives $p = 0.0005$, the smallest value attainable with 2000 permutations.

D.2 Rank-KL variance decomposition

To compare the correlation summaries of Section 3.2, we fit descriptive fixed-effect models to the rank-transformed exact KL in each crossed matrix (Table 4). Receiver identity alone explains most of the rank variance, which is why the within-displacement correlation can be high even though it is not purely pair-specific. Adding displacement identity increases the in-sample additive R^2 further, but 18–25% remains. This remainder is the variation targeted by the residual-rank correlation. Because there is one observed outcome per displacement–receiver cell, we call it non-additive residual variation, not a separately identified interaction.

Table 4: **Descriptive decomposition of rank-transformed exact KL in the six crossed matrices.** "Receiver" and "Displacement" are one-factor fixed-effect R^2 values; "Additive" uses both factors; "Non-additive" is $1 - R^2$ of the additive model. N includes the anchor rows.

Model	Bucket	N	Receiver R^2	Displacement R^2	Additive R^2	Non-additive
Qwen3-8B	B1	8192	0.707	0.123	0.789	0.211
Qwen3-8B	B2	16384	0.733	0.063	0.779	0.221
Qwen3-8B	B3	16384	0.704	0.071	0.754	0.246
Mistral-7B	B1	8192	0.773	0.076	0.804	0.196
Mistral-7B	B2	16384	0.795	0.041	0.815	0.185
Mistral-7B	B3	16384	0.798	0.044	0.820	0.180

D.3 Reconciling the Fisher–KL correlations reported in this paper

The paper reports several Fisher–KL correlations. They differ because they use different estimands, designs, samples, and numbers of receivers per displacement, not because of inconsistent reruns. Table 5 lists them side by side. Three patterns explain the differences. First, within-displacement medians over 8 receivers (control panel) are coarser and lower than those over 15 or 31 receivers (crossed buckets). Second, residual statistics remove the dominant receiver main effect and are therefore lower than within-displacement or pooled ones (Table 4). Third, random subsets of the ordinary matrix (256 dose pairs, 256 wrong-state pairs, 128 teacher-forced pairs) differ from the full 4032-pair value by ordinary sampling variation; at full dose, the dose-subset KL is the same quantity as the ordinary transplant KL.

Table 5: **Every Fisher–KL correlation in the paper, with its design, sample, and estimand.** Ranges are over buckets or depths. All values come from the frozen artifact.

Design and sample	Receivers per Δu	Estimand	Qwen3-8B	Mistral-7B
Crossed B1–B3, primary depth	15 / 31 alternatives	within (primary)	0.858–0.872	0.911–0.928
Crossed B1–B3, primary depth	anchor + 15 / 31	pooled	0.855–0.871	0.921–0.932
Crossed B1–B3, primary depth	anchor + 15 / 31	residual-rank	0.504–0.520	0.658–0.675
Crossed B1, four depths	15 alternatives	within	0.857–0.954	0.854–0.957
Control panel (96 × 8)	8	within	0.738	0.905
Control panel (96 × 8)	8	pooled	0.722	0.876
Control panel (96 × 8)	8	residual-rank	0.456	0.612
Ordinary, all 4032 pairs	—	pooled	0.867	0.911
Ordinary, all 4032 pairs	—	host–donor resid.	0.563	0.677
Ordinary, 256 wrong-state pairs	—	pooled, correct state	0.875	0.919
Ordinary, 256 dose pairs, $\lambda = 1$	—	pooled	0.831	0.907
Ordinary, 128 pairs, $h = 1$	—	pooled	0.857	0.930
Large ordinary, 65,280 pairs	—	pooled / resid.	—	0.911 / 0.660

D.4 Depth sweep

Table 6 gives the B1 crossed-receiver result at every probed depth. The receiver-local score ranks receivers at all four depths in both models, and the within-displacement correlation is positive for at least 99.4% of displacements at every depth.

Table 6: **Crossed-receiver result across depth (bucket B1; 512 displacements, 15 alternative receivers each).** Median within-displacement Spearman correlation between Q_{recv} and exact KL, and the fraction of displacements with a positive correlation.

Model	Layer	Relative depth	Median within $\rho \uparrow$	Positive fraction
Qwen3-8B	7	0.20	0.857	1.000
Qwen3-8B	14	0.40	0.871	0.994
Qwen3-8B	21	0.60	0.936	1.000
Qwen3-8B	26	0.74	0.954	1.000
Mistral-7B	6	0.19	0.854	1.000
Mistral-7B	12	0.39	0.911	1.000
Mistral-7B	19	0.61	0.929	0.994
Mistral-7B	23	0.74	0.957	1.000

D.5 Spread of the effect of a fixed displacement

For each crossed displacement, we compute $\log_{10}(\max_k K_{d,k}/\min_k K_{d,k})$ over its alternative receivers (Table 7). The spread grows with the number of receivers, as any range statistic does, so we compare buckets with the same receiver count.

Table 7: **How much the exact effect of one frozen displacement varies across its receivers.** Median over 512 displacements of $\log_{10}$ of the ratio between the largest and smallest exact KL among the alternative receivers, and the fraction of displacements whose ratio is at least 100.

Model	Bucket	Alternative receivers	Median $\log_{10}$ max/min	Fraction with ratio ≥ 100
Qwen3-8B	B1	15	3.46	0.916
Qwen3-8B	B2	31	4.77	0.994
Qwen3-8B	B3	31	4.36	0.990
Mistral-7B	B1	15	2.90	0.865
Mistral-7B	B2	31	3.31	0.994
Mistral-7B	B3	31	3.65	1.000

D.6 Full alternative-predictor panel

The compact main-text Table 2 reports the primary within-displacement statistic. Table 8 adds the pooled correlation, the fraction of displacements with a positive within-displacement correlation, and the residual-rank correlation. The residual is undefined for the receiver norm, because a predictor that depends only on the receiver is removed entirely by the receiver main effect.

Table 8: **Alternative predictors on the** 96×8 **control panel** (768 rows per model). The magnitudes differ across columns because the estimands differ (Section 3.2).

Model	Predictor	Pooled ρ	Within median ρ	Positive fraction	Residual ρ
Qwen3-8B	Fisher Q_{recv}	0.722	0.738	0.979	0.456
Qwen3-8B	AtP-top	0.512	0.500	0.906	0.283
Qwen3-8B	AtP-margin	0.434	0.429	0.875	0.222
Qwen3-8B	q_{p2}	0.261	0.226	0.708	0.237
Qwen3-8B	$\|\Delta u\|/\|r\|$	0.350	0.071	0.615	0.108
Qwen3-8B	$\|r\|$	-0.081	-0.071	0.354	—
Qwen3-8B	$\lvert\cos(\Delta u, r)\rvert$	-0.056	-0.012	0.479	0.047
Mistral-7B	Fisher Q_{recv}	0.876	0.905	1.000	0.612
Mistral-7B	AtP-top	0.736	0.750	1.000	0.413
Mistral-7B	AtP-margin	0.156	0.179	0.656	0.407
Mistral-7B	q_{p2}	0.089	0.060	0.552	0.197
Mistral-7B	$\|\Delta u\|/\|r\|$	0.118	0.000	0.458	-0.036
Mistral-7B	$\|r\|$	-0.006	0.000	0.479	—
Mistral-7B	$\lvert\cos(\Delta u, r)\rvert$	0.030	0.048	0.531	0.003

D.7 Layerwise causal footprints

Figure 4 shows how the footprints of one displacement at the original host and at a receiver separate across downstream layers. Fidelity and gain both start at one at the intervention site and decay toward zero by the final layer. The receiver's footprint is therefore neither a rescaled copy of the host's nor simply attenuated along the host's direction; most of it lies in a different direction.

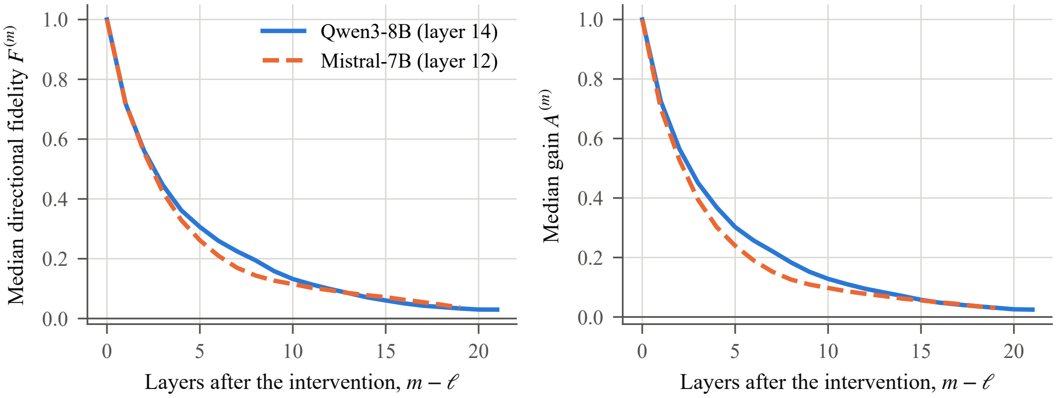

Figure 4: **The downstream direction of a fixed displacement diverges across receivers.** Median over the 768 displacement–receiver pairs of the 96 × 8 control panel of the directional fidelity $F^{(m)}$ (left) and the gain $A^{(m)}$ (right) between the original host's and the receiver's causal footprints, by number of layers after the intervention. At the final layer, median fidelity is 0.030 in Qwen3-8B and 0.037 in Mistral-7B.

D.8 Receiver-specific transformation at the final layer

Table 9 reports the final-layer held-out prediction scores underlying Figure 3. The receiver-specific map improves directional prediction much more than exact vector reconstruction. This supports a modest interpretation: receiver identity contributes a reproducible directional component to the downstream transformation, while substantial update-specific variation remains unexplained.

Table 9: **Held-out final-layer prediction** (Qwen3-8B layer 35, Mistral-7B layer 31; 24 held-out displacements × 8 receivers). Cosine is averaged over receiver–displacement pairs; lower normalized RMSE is better. Bold marks the best predictor per model.

Model	Predictor	Mean cosine ↑	Mean NRMSE ↓
Qwen3-8B	Identity	0.029	1.823
Qwen3-8B	Gain only	0.021	1.000
Qwen3-8B	Global map	0.086	1.005
Qwen3-8B	Receiver-specific map	**0.316**	**0.945**
Qwen3-8B	Wrong-receiver map	−0.023	1.122
Mistral-7B	Identity	0.037	1.859
Mistral-7B	Gain only	0.037	1.000
Mistral-7B	Global map	0.109	0.999
Mistral-7B	Receiver-specific map	**0.241**	**0.986**
Mistral-7B	Wrong-receiver map	0.028	1.080

D.9 Intervention dose

On a random 256-pair subset of the ordinary matrix at the primary depth, we apply $u_i + \lambda \Delta u_{ij}$ for $\lambda \in \{0.10, 0.25, 0.50, 0.75, 1.00\}$ and correlate Q_{recv} with the exact KL at each dose (Figure 5). The correlation is highest at small doses, where the local expansion of Appendix B.1 is most accurate, and it remains high at the full transplant.

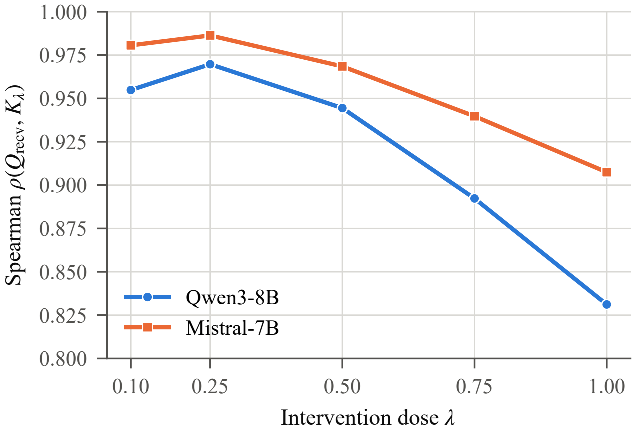

Figure 5: **The local score remains informative as the intervention reaches full dose.** Spearman correlation between Q_{recv} and exact KL at dose λ on 256 ordinary transplant pairs per model at the primary depth. Values at $\lambda = 0.10, 0.25, 0.50, 0.75, 1.00$: $0.955, 0.970, 0.944, 0.892, 0.831$ in Qwen3-8B and $0.981, 0.986, 0.968, 0.940, 0.907$ in Mistral-7B.

D.10 Teacher-forced persistence check

The main experiments measure the next-token distribution. To check whether the association extends beyond the immediate next token, we take a random 128-pair subset of the ordinary matrix at the primary depth and run each patched forward pass on the true C4 continuation. The MLP output is replaced at the final context token only, and later positions see the change only through the model's own computation. We sum the per-position KL over the next $h \in \{1, 4, 8, 16, 32\}$ positions and correlate Q_{recv} with this cumulative disturbance (Table 10 and Figure 6); $h = 1$ is the ordinary next-token KL. Receiver-local sensitivity remains associated with cumulative disturbance beyond the immediate next token, although the association weakens with the horizon, especially in Qwen3-8B. Teacher forcing fixes the continuation, so this check does not establish persistence under unconstrained autoregressive generation.

Table 10: **Teacher-forced persistence.** Spearman correlation between Q_{recv} and KL summed over the next h teacher-forced positions (128 ordinary transplant pairs per model, primary depth).

Model	$h = 1$	$h = 4$	$h = 8$	$h = 16$	$h = 32$
Qwen3-8B	0.857	0.771	0.704	0.614	0.565
Mistral-7B	0.930	0.896	0.839	0.819	0.812

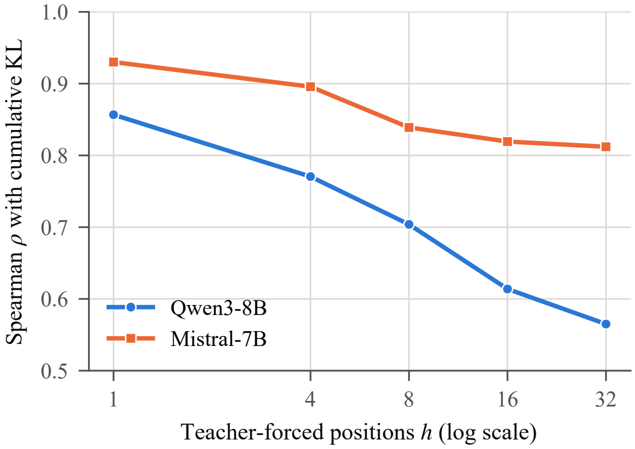

Figure 6: **Association with cumulative disturbance under teacher forcing.** The values of Table 10 plotted against the horizon h (log scale). The association persists beyond the next token and weakens with h.

D.11 Independent-pretraining replication (Pythia-70M, ordinary transplants)

This replication uses ordinary transplants, not the crossed-receiver design. It tests whether the association between local geometry and transplant KL replicates across independently pretrained models. We use the final checkpoints (step 143,000) of Pythia-70M and five further pretraining seeds [Biderman et al., 2023, van der Wal et al., 2025], with MLP layer 2 and FP32 arithmetic. For each run, 256 hosts are drawn from a held-out pool of C4 contexts (corpus B). Each host receives one donor chosen deterministically from its five nearest non-self neighbours among the MLP-input states, after a 32-dimensional PCA. The downstream Fisher score Q_{recv} is computed with exact automatic-differentiation JVPs. The reported statistic is a rank-residualized partial Spearman correlation between $\log(1+Q_{\text{recv}})$ and transplant KL. Both variables are rank-transformed, and the ranks of four controls are regressed out by least squares: the log update distance $\log(1+\|\Delta u\|)$, the log state distance $\log(1+\|\Delta m\|)$ between donor and host MLP inputs, the normalized nonlinear residual $\|\Delta u - J_{F_\ell}(m_i)\Delta m\|/\|\Delta u\|$, and the tangent alignment $\cos(J_{F_\ell}(m_i)\Delta m, \Delta u)$, where J_{F_ℓ} is the exact MLP Jacobian at the host. Table 11 reports the six runs. All six are positive, which shows that the local-geometry association under ordinary transplants is not specific to one pretraining run.

Table 11: **Partial Spearman correlation between** $\log(1+Q_{\text{recv}})$ **and transplant KL on six independently pretrained Pythia-70M runs** (ordinary transplants, layer 2, 256 hosts per run, controls as described in Appendix D.11). The interval is a bootstrap over the six runs.

Run	Hosts	Partial ρ
base	256	0.775
seed1	256	0.824
seed2	256	0.723
seed3	256	0.828
seed4	256	0.762
seed5	256	0.825
Median	—	0.800
95% bootstrap interval	—	[0.742, 0.827]

D.12 Numerical audit of the finite-difference score

The crossed experiments estimate the directional response $JG_\ell(r)\Delta u$ by centered finite differences at $\epsilon = 0.075$ on FP16 model evaluations. We audit this estimate on one frozen 16-pair panel per model: the first 16 control-panel displacements (Appendix C.4), each applied to receiver 0. The 16-pair audit varies displacement direction at a fixed receiver and therefore isolates numerical closure of the local directional derivative; it does not reproduce the primary within-displacement receiver-ranking statistic, which rests on the physical full-dose crossed experiments. We ask two separate questions.

D.12.1 End-to-end FP16 step-size stability

On FP16 model evaluations, we compare the response at ϵ with the response at half step $\epsilon/2 = 0.0375$ and with centered-difference Richardson extrapolation. The median cosine between the responses at ϵ and $\epsilon/2$ is 0.989 for Qwen3-8B and 0.940 for Mistral-7B, and between the ϵ response and the Richardson estimate 0.981 and 0.899. The median relative difference of the Fisher score from the Richardson estimate is 0.333 and 0.472. These end-to-end comparisons combine finite-difference step effects, FP16 finite-precision effects, and the corresponding changes in native logits and probabilities, so they should not be read as truncation error alone. In the ordinary-matrix step-size sweep, rank stability between step sizes is above 0.95 for every comparison involving the chosen $\epsilon = 0.075$; the minimum over all compared pairs of step sizes is 0.949 in Qwen3-8B (for $\epsilon = 0.04$ versus 0.05) and 0.961 in Mistral-7B.

D.12.2 Exact FP32 downstream-suffix closure

To separate truncation error from finite precision, we recompute the same 16 cases with an exact FP32 downstream-suffix JVP from the same cached natural receiver states. The natural states up to and including layer ℓ (the host and donor MLP outputs and the receiver's layer-ℓ output) are produced in FP16, as in the main experiments. The downstream suffix, from layer $\ell + 1$ through the final norm and the language-model head, is then evaluated in FP32 with TF32 disabled. On this suffix, we compute the exact JVP by automatic differentiation and the centered finite differences at ϵ and $\epsilon/2$, with their Richardson combination. The upstream computation is not repeated in FP32, so this is not a full-model FP32 JVP. The host, donor, and receiver context identifiers match the frozen control panel for all 16 cases in both models.

Table 12: **Numerical audit on 16 frozen cases per model** (control-panel displacements 0–15 at receiver 0; Qwen3-8B layer 14, Mistral-7B layer 12). The upper block compares FP32 finite differences with exact FP32 suffix JVPs. The lower block compares the FP16 finite-difference scores for these same 16 control-panel cases, computed with the same estimator used in the crossed experiments, with the exact FP32 suffix-JVP scores. "Relative Q error" is $|Q - Q_{\mathrm{JVP32}}|/Q_{\mathrm{JVP32}}$.

Comparison	Statistic	Qwen3-8B	Mistral-7B
FD32 ($\epsilon = 0.075$) vs. JVP32	median directional cosine	1.0000	1.0000
	Spearman $\rho(Q_{\mathrm{FD32}}, Q_{\mathrm{JVP32}})$	1.000	1.000
	median relative Q error	0.21%	0.20%
	maximum relative Q error	0.82%	0.37%
FD32 ($\epsilon/2$) vs. JVP32	median relative Q error	0.12%	0.07%
FD32 Richardson vs. JVP32	median relative Q error	0.18%	0.13%
FP16 FD ($\epsilon = 0.075$) vs. JVP32	Spearman $\rho(Q_{\mathrm{FP16}}, Q_{\mathrm{JVP32}})$	0.785	0.544
	median relative Q difference	21.6%	26.7%

Table 12 reports the result. In FP32, the centered finite difference at $\epsilon = 0.075$ reproduces the exact JVP almost perfectly: every directional cosine is at least 0.99998, the Fisher scores are identically ranked, and the median relative score error is about 0.2% in both models, with the half step and Richardson combination equally close. The step $\epsilon = 0.075$ itself therefore introduces negligible centered-difference truncation error on this panel. The FP16 finite-difference scores for the same 16 cases agree less closely with the exact FP32 scores (Spearman 0.785 and 0.544; median relative difference 21.6% and 26.7%), and Richardson extrapolation of the FP16 differences moves further from them (Spearman 0.697 and 0.474), consistent with amplification of finite-precision noise. On this audit panel, this pattern is consistent with finite-precision effects dominating the observed discrepancy in Appendix D.12.1 rather than centered-difference truncation error. Accordingly, we treat the large-scale FP16 Fisher values as ranking diagnostics rather than as numerically exact FP32 quadratic forms; the audit closes truncation error on the tested directions but does not make the FP16 scores exact. The causal-footprint and held-out receiver-map results in Sections 5.5 and 5.6 do not depend on the finite-difference Fisher estimator.

E Scope and Interpretation

The crossed design supports a causal statement about the receiving state, because the intervention vector is physically held fixed within each displacement. It does not establish that an MLP update represents a context-independent semantic function, and we make no such claim. Likewise, the low-rank maps should not be read as complete or deterministic receiver operators. Their held-out cosines are meaningful but far from one, and their normalized RMSE gains are modest. The result is better read as evidence that a reproducible receiver-specific component coexists with substantial update-specific variation.

The next-token endpoint is deliberately narrow. It gives a precise, full-vocabulary behavioral measure and makes exact physical controls possible, but it does not characterize free-running generation. The paper is diagnostic and proposes no steering or editing method. Natural extensions include longer-horizon behavior, other component types such as attention outputs, mechanistic decomposition of the receiver dependence, and intervention methods that aim to preserve a known causal effect across receiver states.

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Justification: The physical crossed-receiver study was run in a Kaggle session with two NVIDIA Tesla T4 GPUs (16 GB VRAM each). Models were loaded in FP16 with Accelerate sharding (`device_map="balanced"`, 12,000 MiB memory cap per GPU), gradients disabled, 64-token contexts, and pair batch 4. The preserved run contains 12 crossed-receiver stages—four bucket-1 depths for each model plus bucket-2 and bucket-3 primary-depth replications—for 131,072 physical crossed-receiver interventions in total, together with a 65,280-pair Mistral full-core stage. Instrumented successful-stage runtimes in `ALL_STAGE_SUMMARIES.csv` sum to 631.3 minutes (10.52 hours wall-clock): 437.8 minutes for the crossed-receiver stages and 193.4 minutes for the Mistral full-core stage. Because both T4s shard each large model, this corresponds to roughly 21 T4 GPU-hours if both devices are counted for the full measured interval. The observed end-to-end successful notebook run was approximately 10–12 hours, including model loading, native-state caching, and ordinary notebook/session overhead. The eight ordinary 64-context depth targets ran on the same hardware class and total 134.9 minutes of recorded target runtime; the two 96×8 control-panel runs used the same hardware class, but their wall-clock time was not instrumented, so we do not report it. These figures exclude exploratory development and failed engineering attempts, which used additional compute. The saved row-level artifact can be reanalyzed on CPU without rerunning the GPU experiments (Appendix C.6). The exact FP32 downstream-suffix JVP audit (Appendix D.12.2; 16 pairs per model) also ran on Kaggle with two NVIDIA T4 GPUs (16 GB each): natural states were produced with the FP16 model, and the downstream suffix was evaluated in FP32 with TF32 disabled, split across both GPUs. Its runtime was not recorded, so we do not report it.

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